

1. Process Creation – Process ID and Parent Process ID

```
import threading

counter = 0
mutex = threading.Lock()

def increment():
    global counter

    for i in range(100000):
        with mutex:
            counter += 1

t1 = threading.Thread(target=increment)
t2 = threading.Thread(target=increment)

t1.start()
t2.start()

t1.join()
t2.join()

print("Final Counter Value:", counter)
```

OUTPUT

```
= RESTART: C:/Users/ATHIN/AppData/Local/Programs/Python/Python39-64/Python.exe
ssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssss
Parent Process
Parent Process ID: 26620

Process completed
|
```

2. Copy One File to Another

```
source = input("Enter source file name: ")
destination = input("Enter destination file name: ")

try:
    with open(source, "r") as file1:
        data = file1.read()

    with open(destination, "w") as file2:
        file2.write(data)

    print("File copied successfully")

except FileNotFoundError:
    print("Source file not found")
```

OUTPUT

```
= RESTART: C:/Users/ATHIN/AppData/Local/Prog  
ssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssss  
Enter source file name: athin  
Enter destination file name: athinn  
Source file not found
```

3. FCFS Scheduling


```

n = int(input("Enter number of processes: "))

processes = []

for i in range(n):
    bt = int(input("Enter burst time for P" + str(i + 1) + ": "))
    processes.append([i + 1, bt])

processes.sort(key=lambda x: x[1])

waiting = 0
total_wt = 0
total_tat = 0

print("\nProcess\tBT\tWT\tTAT")

for p in processes:
    pid = p[0]
    bt = p[1]

    tat = waiting + bt

    print("P" + str(pid), "\t", bt, "\t", waiting, "\t", tat)

    total_wt += waiting
    total_tat += tat
    waiting += bt

print("\nAverage Waiting Time =", total_wt / n)
print("Average Turnaround Time =", total_tat / n)

```

output

```

#####
Enter number of processes: 5
Enter burst time for P1: 23
Enter burst time for P2: 32
Enter burst time for P3: 22
Enter burst time for P4: 33
Enter burst time for P5: 45

Process BT      WT      TAT
P3      22      0       22
P1      23      22      45
P2      32      45      77
P4      33      77      110
P5      45      110     155

Average Waiting Time = 50.8
Average Turnaround Time = 81.8

```

5. Priority Scheduling


```

n = int(input("Enter number of processes: "))

at = []
bt = []
priority = []

for i in range(n):
    print("\nProcess P" + str(i + 1))

    at.append(int(input("Enter arrival time: ")))
    bt.append(int(input("Enter burst time: ")))
    priority.append(int(input("Enter priority: ")))

remaining = bt.copy()
ct = [0] * n

time = 0
completed = 0

while completed < n:

    selected = -1
    highest = 999999

    for i in range(n):
        if at[i] <= time and remaining[i] > 0:
            if priority[i] < highest:
                highest = priority[i]
                selected = i

    if selected == -1:
        time += 1
        continue

    remaining[selected] -= 1
    time += 1

    if remaining[selected] == 0:
        ct[selected] = time
        completed += 1

```

OUTPUT

```

= RESTART: C:/Users/ATHIN/AppData/Local/Programs/Python/Pyt
ssssssssssssssssssssssssssssssssssssssssssssssssssssssssssssss.py
Enter number of processes: 4

Process P1
Enter arrival time: 1
Enter burst time: 23
Enter priority: 2

Process P2
Enter arrival time: 2
Enter burst time: 23
Enter priority: 1

Process P3
Enter arrival time: 4
Enter burst time: 34
Enter priority: 3

Process P4
Enter arrival time: 8
Enter burst time: 455
Enter priority: 5

Process AT      BT      Priority      WT      TAT
P1              1       23        2        23       46
P2              2       23        1         0       23
P3              4       34        3        43       77
P4              8      455        5        73      528

```

7. Non-Preemptive SJF


```

from multiprocessing import Process, Value

def child(shared_value):

    print("Child received value:", shared_value.value)

    shared_value.value = shared_value.value + 10

    print("Child changed value to:", shared_value.value)

if __name__ == "__main__":

    shared_value = Value("i", 20)

    print("Parent initial value:", shared_value.value)

    p = Process(target=child, args=(shared_value,))

    p.start()
    p.join()

    print("Parent final value:", shared_value.value)

```

OUTPUT

```

= RESTART: C:/Users/ATHIN/App
ssssssssssssssssssssssssssssssssssssss
Parent initial value: 20
Parent final value: 30

```

10.Message Queue

```
from multiprocessing import Process, Queue

def sender(q):

    message = "Hello from Sender Process"

    q.put(message)

    print("Message Sent:", message)

def receiver(q):

    message = q.get()

    print("Message Received:", message)

if __name__ == "__main__":

    q = Queue()

    p1 = Process(target=sender, args=(q,))
    p2 = Process(target=receiver, args=(q,))

    p1.start()
    p1.join()

    p2.start()
    p2.join()

    print("Communication completed")
```

OUTPUT

[illegible]